BASE CAMP

The Roadmap at a Glance

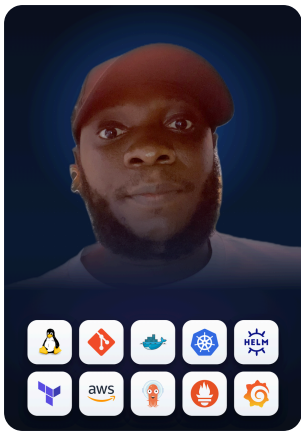
DevOps Roadmap: Beginner to Senior

A practical learning path for becoming a Senior DevOps / Cloud / Platform Engineer.

Your instructor: Ubong Eduok

Author and instructor - DevOps Roadmap: Beginner to Senior

GitHub: @mascot200 `(github.com/mascot200)`



This repository turns the roadmap into a **hands-on curriculum**. Every level is a folder. Every topic inside a level is one page that contains:

- a plain-English explanation and **why it matters** in production,
- a **real-world system example** (how companies and well-known open-source systems use it),
- **Mermaid diagrams** (rendered natively on GitHub),
- a **step-by-step guide** you can copy-paste in a sandbox,
- a realistic **scenario**, common mistakes and best practices,
- **4-6 tasks** with verifiable "done when" checks, and self-check questions.

Each level ends with a **capstone project**, and nine larger projects chain the levels into one evolving production system.

The lifecycle

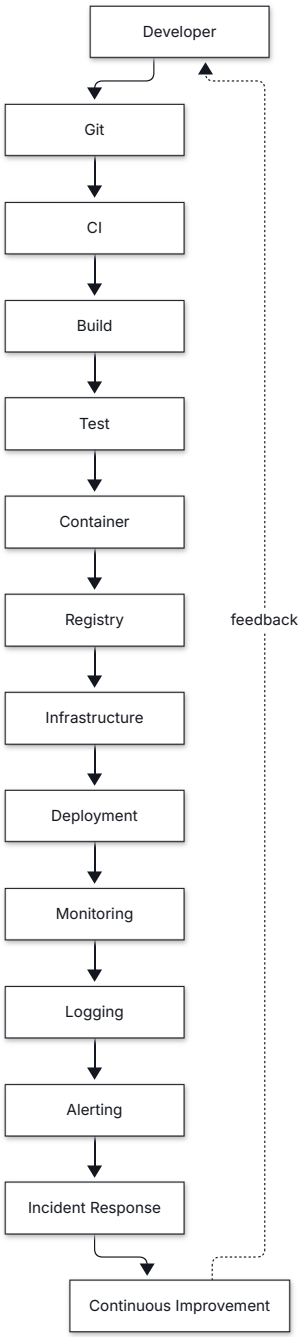


Figure 1

Levels

Level	Title	Time	Topics	Capstone
0	What DevOps Actually Is Level 0	1 week (reading + mapping)	13	Capstone Capstone 0
1	Linux Fundamentals Level 1	2-4 weeks	13	Capstone Capstone 1
2	Networking Level 2	2-4 weeks	11	Capstone Capstone 2
3	Git & GitHub Level 3	2-3 weeks	13	Capstone Capstone 3
4	Docker Level 4	2-3 weeks	14	Capstone Capstone 4
5	CI/CD Level 5	2-3 weeks	12	Capstone Capstone 5
6	Cloud (AWS) Level 6	4-6 weeks	18	Capstone Capstone 6
7	Terraform / Infrastructure as Code Level 7	2-3 weeks	9	Capstone Capstone 7
8	Kubernetes Level 8	6-10 weeks	19	Capstone Capstone 8
9	Helm Level 9	1-2 weeks	6	Capstone Capstone 9
10	Observability Level 10	3-4 weeks	11	Capstone Capstone 10
11	Security / DevSecOps Level 11	3-4 weeks	13	Capstone Capstone 11
12	Advanced Networking Level 12	3-4 weeks	14	Capstone Capstone 12
13	Databases & Distributed Systems Level 13	4-6 weeks	15	Capstone Capstone 13
14	SRE Level 14	3-4 weeks	10	Capstone Capstone 14
15	Disaster Recovery Level 15	2-3 weeks	10	Capstone Capstone 15
16	Advanced Kubernetes Level 16	4-6 weeks	10	Capstone Capstone 16
17	GitOps Level 17	2-3 weeks	7	Capstone Capstone 17
18	Platform Engineering Level 18	4-6 weeks	10	Capstone Capstone 18
19	Architecture Level 19	Ongoing	11	Capstone Capstone 19
20	Cost Optimization Level 20	2-3 weeks	9	Capstone Capstone 20

Guides

- Recommended learning order
- Suggested timeline
- The senior-level mindset
- Practical project progression

Deployment agents and skills (Claude Code)

The roadmap teaches you to deploy by hand so you understand why every step exists. When you are ready to **ship real applications with an AI pair-deployer**, use the companion toolkit: `devops-deploy-toolkit` `(github.com/mascot200/devops-deploy-toolkit)` - Claude Code agents, skills and scripts that deploy to a VPS, AWS, Kubernetes or a GitOps cluster with one rule built into the code: **look first, change only when a human has approved the plan.**

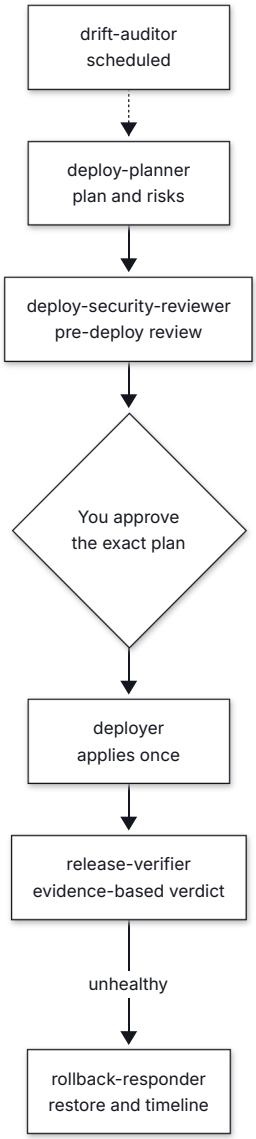


Figure 2

Target	Skill	Roadmap lessons it builds on
Docker Compose on a VPS	`deploy-compose-vps`	Docker Compose §4.8, Rollback §5.7
AWS with Terraform	`deploy-terraform-aws`	Terraform Level 7, Drift detection §7.9
Kubernetes with Helm	`deploy-helm-k8s`	Helm Level 9, Probes §8.15
GitOps with Argo CD	`deploy-gitops-argocd`	GitOps Level 17, Rollback through Git §17.6
Any of the above	`deploy-preflight`, `post-deploy-verify`	CI/CD pipeline design §5.1, What to monitor §10.11

```bash
git clone https://github.com/mascot200/devops-deploy-toolkit.git
cd your-app-repo && ../devops-deploy-toolkit/install.sh --dry-run   # then without --dry-run
```

Everything is dry-run by default, production needs an explicit confirmation, and floating tags such as `latest` are refused. The scripts are tested with stub binaries and real local git; they have not been run against a live cloud account, so rehearse in staging first. Full guide: `docs/GUIDE.md` (github.com/mascot200/devops-deploy-toolkit/blob/main/docs/GUIDE.md).

Recommended path (summary)

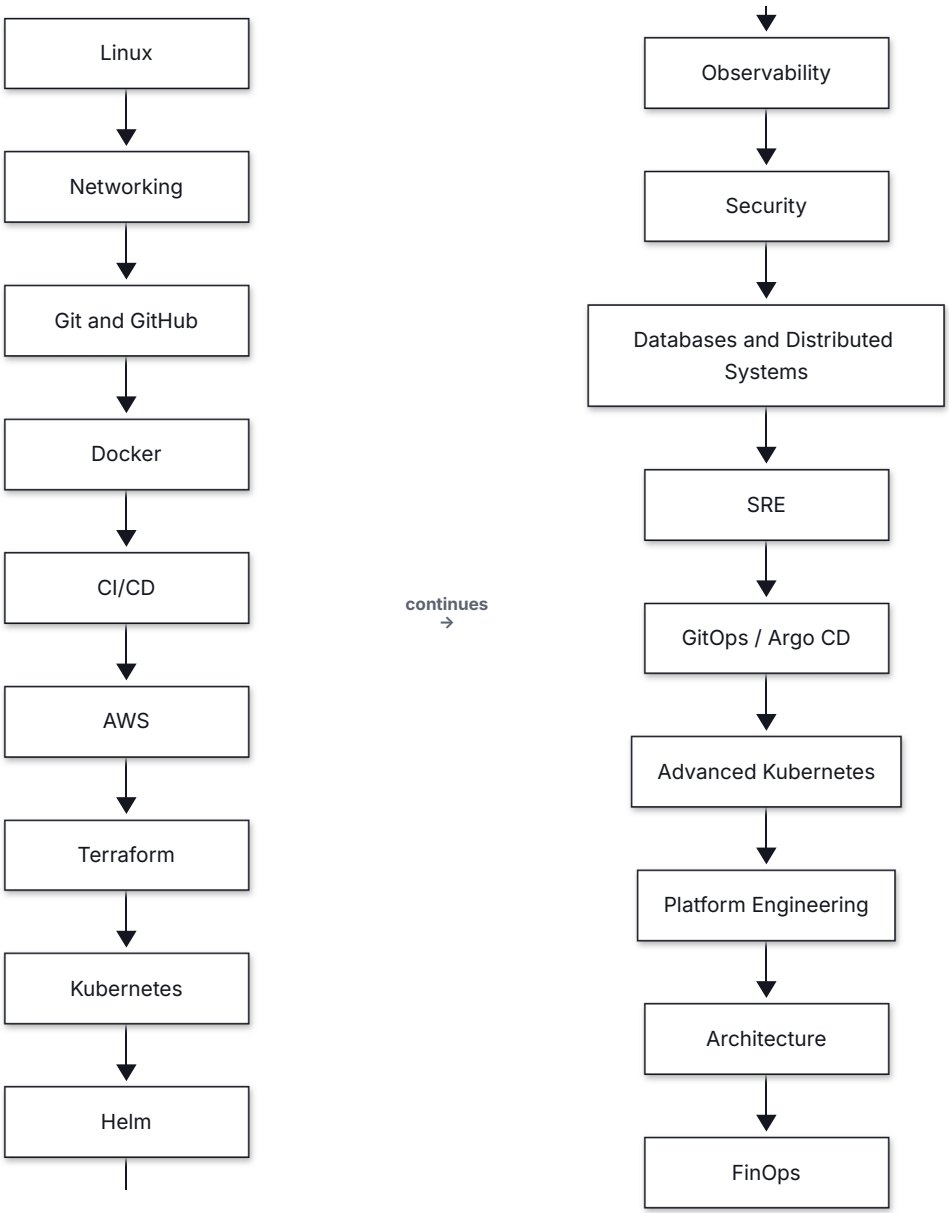


Figure 3

> Level numbers follow the roadmap document; the *learning order* above is the recommended sequence (for example Security and Databases come before SRE, and Advanced Kubernetes

> after GitOps). See the learning-order guide.

How to use this repository

1. Fork it (or copy a level into your own notes repo) so you can tick off tasks: change `- [ ]` to `- [x]`.
2. Work through one **running project** across the whole journey. The sample app used in all examples is `shop-api` (Node.js API + PostgreSQL + Redis + Nginx).
3. Use disposable sandboxes: local VMs, Docker, kind/minikube, LocalStack, or a small VPS. Anything that can cost money says so and ends with teardown steps.
4. Keep a *lab notebook*: for each task, write what you ran and what you saw. Explaining output is the skill.

Repository layout

```
devops-roadmap/
├── README.md
├── docs/TOPIC_TEMPLATE.md          # the structure every topic follows
├── docs/images/                    # instructor photo and other images
├── guides/                         # learning order, timeline, senior mindset
├── projects/                       # nine end-to-end projects
├── level-00-what-devops-is/
│   ├── README.md                   # level index and goals
│   ├── 01-sdlc.md                  # one file per topic
│   └── 99-capstone-project.md      # level capstone
├── level-01-linux-fundamentals/
└── ... level-20-cost-optimization/
```

Senior-level mindset

Do not aim to become the person who knows 50 DevOps tools. Aim to become the engineer who can take a software application and independently make it production-ready: **secure, observable, scalable, highly available, automatically deployable, recoverable, and cost-efficient.**

License

Released under the MIT License Appendix D - Copyright (c) 2026 Ubong Eduok. You may use, copy, modify and share this material, including in your own courses and projects, as long as the copyright and license notice are kept. Everything is provided "as is"; run labs in sandboxes and read cost warnings before using real cloud accounts.